



# Straight lines

A STRAIGHT LINE IS USUALLY JUST CALLED A LINE. IT IS THE SHORTEST DISTANCE BETWEEN TWO POINTS ON A SURFACE OR IN SPACE.

## SEE ALSO

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◀ 84–85 Angles

Constructions 110–113 ▶

## Points, lines, and planes

The most fundamental objects in geometry are points, lines, and planes. A point represents a specific position and has no width, height, or length. A line is one dimensional—it has infinite length extending in two opposite directions. A plane is a two-dimensional flat surface extending in all directions.

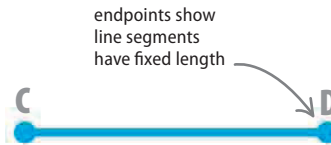
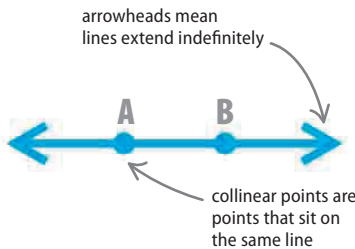
A •

• B

• C

### △ Points

A point is used to represent a precise location. It is represented by a dot and named with a capital letter.



### △ Lines

A line is represented by a straight line and arrowheads signify that it extends indefinitely in both directions. It can be named by any two points that it passes through—this line is AB.

### △ Line segments

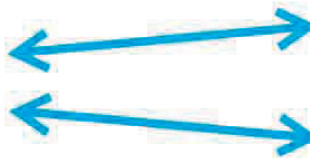
A line segment has fixed length, so it will have endpoints rather than arrowheads. A line segment is named by its endpoints—this is line segment CD.

### △ Planes

A plane is usually represented by a two-dimensional figure and labeled with a capital letter. Edges can be drawn, although a plane actually extends indefinitely in all directions.

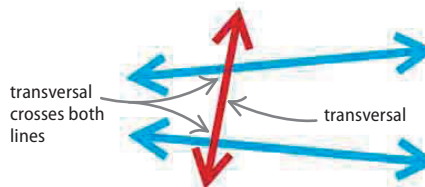
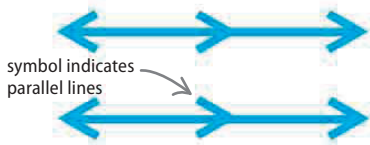
## Sets of lines

Two lines on the same surface, or plane, can either intersect—meaning they share a point—or they can be parallel. If two lines are the same distance apart along their lengths and never intersect, they are parallel.



### △ Nonparallel lines

Nonparallel lines are not the same distance apart all the way along; if they are extended they will eventually meet in a point.



### △ Parallel lines

Parallel lines are two or more lines that never meet, even if extended. Identical arrows are used to indicate lines that are parallel.

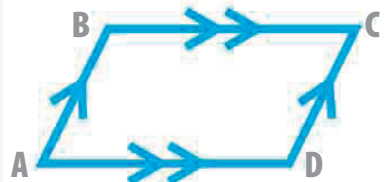
### △ Transversal

Any line that intersects two or more other lines, each at a different point, is called a transversal.

## LOOKING CLOSER

### Parallelograms

A parallelogram is a four-sided shape with two pairs of opposite sides, both parallel and of equal length.



### △ Parallel sides

The sides AB and DC are parallel, as are sides BC and AD. The sides AB and BC, and AD and CD are not parallel—shown by the different arrows on these lines.